The empirical recurrence for OEIS A253355, recovered from its tail

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Abstract

OEIS A253355 records "Empirical recurrence of order 43" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 2470$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 48$. Its last coefficient is $c_{43} = -530404409239142400 \neq 0$, so no further tail satisfies anything shorter; any order-43 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A253355 is "Number of (6+1)X(n+1) 0..1 arrays with every 2X2 subblock diagonal maximum minus antidiagonal maximum nondecreasing horizontally, vertically and ne-to-sw antidiagonally.". It has offset 1 and begins

$$2470, 60779, 1285960, 29779356, 664592224, 15101198444, 340810754556, 7718859534172, \ \ldots$$

The entry states:

> Empirical recurrence of order 43 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 13:59:32, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 2470$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 2470$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a. ☐

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 43: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 43.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 4940$ exact terms from $n = 5$ on, it returns order 43 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{43} c_i\, a(n-i),$$

c_1	72	c_{12}	675698711376	c_{23}	2135885372576268288	c_{34}	139601887625434103808
c_2	−1931	c_{13}	16578796382048	c_{24}	−2011581611939921920	c_{35}	−11922098839099120025
c_3	19538	c_{14}	−64756353344576	c_{25}	−7584501613718536192	c_{36}	−7862928611843715891
c_4	100413	c_{15}	−134173857715072	c_{26}	13784598207526076416	c_{37}	84845767721914204160
c_5	−4242004	c_{16}	1349514026669568	c_{27}	15135741320257601536	c_{38}	25842267261627269120
c_6	31231687	c_{17}	−1039438519612416	c_{28}	−49130653604522229760	c_{39}	−3410718618209694515
c_7	110702986	c_{18}	−13947460019884032	c_{29}	−9105973595592458240	c_{40}	−4415342928520544256
c_8	−2919035988	c_{19}	34445222660911104	c_{30}	10843068496458285056 0	c_{41}	6996913767166509056
c_9	11380887064	c_{20}	70739200089346048	c_{31}	−32076543208607186944	c_{42}	301538864894312448
c_{10}	71438254316	c_{21}	−362446115954745344	c_{32}	−153800125421683998720	c_{43}	−530404409239142400
c_{11}	−735184874216	c_{22}	−24525620663205888	c_{33}	9383222041174133964 8		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 48$, and it is the recurrence OEIS A253355 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{43} - c_1 x^{43-1} - \cdots - c_{43}$. Its constant term is $-c_{43} = 530404409239142400$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 43 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 43, so $\deg q = 43$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 43, so $q = q_{\min}$. ☐

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 14 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 43, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -530404409239142400.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-43 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A253355.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)

[5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.